Quantitative Detection and Classification of Acoustic Cavitation Using Wavelet Packet Transform and k-Means Clustering

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Abstract

Acoustic emissions from cavitation bubbles contain complex information in the time and frequency domains. Passive cavitation detectors are used to measure these emissions. By processing this data, it becomes possible to extract key cavitation features such as broadband emissions, harmonics, sub- and ultra-harmonics. This allows for cavitation monitoring which has a wide range of industrial applications. This paper presents a novel approach that utilizes the wavelet packet transform, statistical metrics and k-means clustering to detect, quantify and classify cavitation features in acoustic emissions. Additionally, the algorithm quantifies the frequency bandwidth of the broadband component, associated with transient cavitation. The findings enable us to identify and quantitatively characterise the periodic and transient features of acoustic cavitation. The method is validated using synthetic acoustic emissions generated by solving the single bubble dynamics (Gilmore-Akulichev model) and is further tested on experimental data from biomedical ultrasound applications.

ACKNOWLEDGMENTS

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I. INTRODUCTION

Acoustic cavitation refers to the formation, growth and oscillation of bubbles (cavities filled with gas and vapour) in a liquid using acoustic waves. [1] These bubbles may periodically oscillate whilst exposed to acoustic waves (known as stable cavitation) or grow and implode violently within a few cycles of excitation (known as inertial/transient cavitation). There are several physical effects associated with acoustic cavitation, including creation of free radicals, shock waves, shear stresses, micro-jets and micro-streaming in the insonated liquid.[1] To utilize these effects for different applications, ultrasound waves are widely used to produce acoustic cavitation, e.g. imaging ultrasound,[2, 3] therapeutic ultrasound, [4–6] sono-chemistry,[7, 8] and sono-crystallization.[9–12] To monitor these processes

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and to perform mechanistic studies, acoustic cavitation must be measured and evaluated quantitatively.

Passive[5, 13] and active [14, 15] cavitation detection are the two main techniques used in biomedical ultrasound to quantitatively measure acoustic emissions generated by bubbles. Passive techniques use a single element ultrasound transducer, i.e. passive cavitation detector (PCD), to acquire acoustic emissions whereas active techniques use more sophisticated ultrasound array transducers. Passive methods are cost-effective and easy to implement, hence they have been widely used in basic and applied research. Some examples in biomedical ultrasound are as follows: hyperthermia and thermal ablation,[16] histotripsy,[17] drug and gene delivery,[18] blood-brain-barrier opening.[19, 20] Several methods have been developed to analyze acoustic emissions from cavitations, including spectral techniques, combination of spectral and digital filters and/or singular-value decomposition, time-frequency methods including short time-Fourier transform, Bilinear time-frequency distributions, and wavelet transforms.[5, 21–25] These works detect and classify the two main types of acoustic cavitation qualitatively. To carry out fundamental studies and elucidate the concomitant biological/physical effects produced by acoustic cavitation, advanced signal processing techniques are required to detect, extract, classify and quantitatively characterize the features of acoustic cavitation. To tackle this problem, a method was developed based on discrete wavelet decomposition and spectral and statistical analyses to identify different types of cavitation and estimate their energies.[26] Due to the limitations of the discrete wavelet transform, this method cannot isolate cavitation features in complex scenarios with overlapping spectral signatures. In this paper, a new method is presented which uses combined discrete wavelet packet transform, statistical signal processing and k-means clustering to quantitatively analyze acoustic emissions by extracting and classifying cavitation features. The method is implemented in a Python library which is available in a GitHub repository. [27]

The method is discussed in detail in Section II. To test and verify the accuracy and efficiency of the method, synthetic acoustic emissions were generated by solving the single bubble dynamics problem using the Gilmore-Akulichev model, presented in Section II F. The method was then used to analyze the theoretical acoustic emissions. Finally, the method was used to process experimental PCD data, explained in Section III C.

II. MATERIALS AND METHODS

A. Features of acoustic cavitation

The signal from a PCD may exhibit multiple spectral features, influenced by acoustic parameters and the physical properties of the liquid: (i) harmonics at frequencies of nf where f is the fundamental (excitation) frequency and n is a positive integer, (ii) sub-harmonics at frequencies of nf/m where $m > n$ is an integer that is not divisible by n, (iii) ultra-harmonics at frequencies of mf/n, (iv) a broadband frequency component. The relationship between bubble dynamics and the these spectral features is complex, with various hypotheses proposed and studied in literature. [28–33].

The fundamental frequency f is the main component observed when a bubble undergoes low-amplitude periodic oscillations or behaves as a scatterer, such as when the initial radii are larger than the incident wavelength. [30] In the stable cavitation regime, e.g. at low intensities, a bubble may begin to oscillate nonlinearly due to the nonlinear coupling between free and forced pulsations. This process emits high-order harmonics, with the second harmonic $2f$ often being the most prominent. The amplitude of the second harmonic is proportional to the square of the fundamental. [30] Theoretical studies show that for these nonlinear oscillations to occur, the ratio of the excitation frequency to the linear resonance frequency of the bubble, denoted by f_r, must be a rational fraction, i.e. $f/f_r = n/m$ and the pressure amplitude must be above a certain threshold. [30] The emissions from these nonlinear stable pulsations appear as sub-harmonics if $m = 1$, meaning that the bubbles are driven at integer multiples of their resonances. Harmonics are produced if $n = 1$ and ultra-harmonics are produced otherwise (i.e., $n \neq m$). [28, 31, 33] A bubble excited by a pulsed excitation with a small duty cycle and a large repetition period may also emit low-energy signals at the $f/2$ sub-harmonic. [30]

In the transient cavitation regime, at intensities above the transient cavitation threshold, the amplitudes of spectral features increase. The sub-harmonic signals are explained by the prolonged expansion phase that can occur when a bubble grows or oscillates over several cycles of sonication before collapsing. Additionally, a bubble with an initial size whose resonance frequency is close to sub-harmonics may resonate and undergo transient cavitation. The chaotic dynamics of transient bubbles [33], and shocks produced by inertial collapses

manifest as a broadband component in the spectrum, which is often considered a hallmark of transient cavitation. [22] In the case of a cloud of bubbles, the interaction between bubbles can produce harmonics and sub- and ultra-harmonics as well [34], further complicating the analysis of the PCD data. Moreover, emitted shocks can set neighboring bubbles into oscillations at their resonance frequencies which may appear as spectral contents unrelated to the excitation frequency f.

These spectral features of acoustic emissions exhibit distinct time-frequency structures depending on the cavitation regime. In stable cavitation, the (sub- and ultra-) harmonic components of the signal may persist over extended time intervals with localized frequency values corresponding to mf/n. In contrast, during transient cavitation, the emissions display impulsive temporal structures. These transient impulsive components can manifest in the frequency domain as a broadband component and localized spectral components at mf/n.

The noise component (primarily measurement noise) exists over a long period and manifests as a broadband distribution in the frequency domain. Separating measurement noise from the cavitation broadband component is challenging as their statistical features are unknown. Additionally, the bandwidth of the cavitation broadband component is usually identified arbitrarily. To quantify these multi-resolution features of the spectrum and determine the bandwidth and energy of cavitation broadband component, a method is proposed that utilizes the wavelet packet transform and k-means clustering, which is discussed in the following section.

B. Wavelet packet transform

Wavelet packet bases were developed to effectively decompose a piecewise smooth signal with a varying time-frequency structure. Orthonormal wavelet packet bases divide the frequency axis into separate intervals of various sizes, known as scales.

In the fast orthogonal discrete wavelet transform (DWT), a bank of conjugate mirror filters and downsampling (by a factor of 2) are used to decompose a signal $x[n]$ of length $N = 2^{n_0}$ into a cascade of coarser approximations $a_j[n]$ and finer details $d_j[n]$ at a scale j [35]

$$
\begin{aligned}
a_{j+1}[n] &= \sum_{m=0}^{M-1} h[m] a_j[m-2n] \\
d_{j+1}[n] &= \sum_{m=0}^{M-1} g[m] a_j[m-2n]
\end{aligned}
\tag{1}
$$

where $a_0[n] = x[n]$, $h[m]$ and $g[m]$ are the impulse responses of the filters corresponding to the scaling and wavelet bases functions and are related as follows

$$
g[m] = (-1)^{1-m} h[1-m] \tag{2}
$$

$g[m]$ and $h[m]$ can be viewed as transfer functions of low- and high-pass filters, respectively, with length M. Thus, approximations and details represent the amplitudes of low- and high- frequency components of the decomposed signal respectively. In Eq. (1) , only the approximations (a_j) are divided to low- and high-frequency components. In the DWPT, both approximation (a_j) and detail (d_j) are split at each scale, represented as a binary tree. This increases the computational complexity of the discrete wavelet packet transform (DWPT) to $O(Nlog_2(N))$ compared to $O(N)$ for the DWT. Each node in the binary tree is referred to as a wavelet packet and includes a set of coefficients from the transform corresponding to a frequency subband. Nodes are labeled with a tuple (j, k) where j is the scale number (the depth of the node in the tree) and k is the number of nodes that are on its left at the same scale.

The properties of the high- and low-pass filters are determined by the family and order of the wavelet function. There is a trade-off between the time and frequency resolution when selecting the order of the wavelet. A higher-order wavelet leads to subbands with less overlap, due to sharper cutoff and higher stopband attenuation, but it is computationally expensive and sacrifices in time resolution.

With more levels of decomposition, the frequency subbands become narrower. To determine the final decomposition levels for each signal, the Shannon entropy minimization algorithm was used. [36] In this approach, if the combined entropy of the child nodes exceeds that of the parent node, the nodes are considered to contain more informative data and are retained in the transform tree. This entropy-based method tends to further decompose strong harmonics while preserving transient features at higher-level nodes, reducing

redundancy in the wavelet packet transform and providing a more compact signal representation. It also addresses class imbalance caused by the excessive number of nodes at higher frequencies in the full DWPT, making efficient use of DWPT's flexibility in tiling the time-frequency domain.

The wavelet packets can be separately reconstructed into signals which combine to form the original signal. This enables the transform to separate out stable and transient components of the original acoustic emission, which tend to have different time-frequency signatures. Once separated, the packets can be classified to indicate the presence of different cavitation features and quantify them.

C. Statistical metrics for feature detection

To extract the signatures of inertial and stable cavitation in each subband, the DWPT coefficients of the leaf nodes, i.e., the nodes with no further decomposition, are characterized using the following statistical metrics: energy, kurtosis, crest factor, skewness, entropy and the logarithm of energy. The energy of a packet, which measures how much of the signal is concentrated within a given subband, is calculated as follows

$$E_{j,k} = \sum_{n=1}^{N_{j,k}} x_{j,k}[n]^2 \tag{3}$$

where $x_{j,k}[n]$ are coefficients in a subband (j,k) and $N_{j,k}$ is the number of coefficients in the subband. $E_{0,0}$ is the total energy of the original signal. Note that using orthogonal wavelet bases for DWPT preserves the energy of the signal. Hence, the sum of the energy of all nodes in the final scale equals to the energy of the time series signal. The average energy of the coefficients in a node is defined as

$$\bar{E}_{j,k} = \frac{1}{N_{j,k}} E_{j,k} \tag{4}$$

where $E_{j,k}$ is the energy of the coefficients in a subband (j,k) and $N_{j,k}$ is the number of coefficients in the subband.

To better distinguish fundamental, sub- and ultra-harmonics, the log-energy metric is also used, i.e., $LE_{j,k} = \log E_{j,k}$. To measure the energy distribution of a subband - compared to other subbands, Shannon entropy is used, defined as follows

$$S_{j,k} = -\sum_{n=1}^{N_{j,k}} \frac{x_{j,k}[n]^2}{E_{j,k}} \log \frac{x_{j,k}[n]^2}{E_{j,k}}. \tag{5}$$

Level entropy is a variation on the calculation of entropy given by

$$S'_{j,k} = -\sum_{n=1}^{N_{j,k}} \frac{x_{j,k}[n]^2}{E_{0,0}} \log \frac{x_{j,k}[n]^2}{E_{0,0}} \tag{6}$$

Level entropy is also used to obtain the minimum entropy decomposition for DWPT.

The normalized spectral entropy $S(f)_{j,k}$ is a measure of the complexity of the distribution in the spectrum of the subband. This is calculated using Eq. (5) where $x_{j,k}[n]$ is replaced with its magnitude spectrum and normalized by $\log(N_{j,k})$. The normalized spectral entropy is in [0,1], with the lower the normalized spectral entropy of a subband, the more deterministic components (information about the fundamental and harmonics) are within the subband. Subbands with flat power spectral density have values close to 1, corresponding to the highest possible entropy which is a characteristic of white noise. The transient components have a broadband frequency signature, hence a higher entropy can be attributed to transient and/or noise components of the signal. To distinguish these features additional statistical metrics are required.

Kurtosis measures the occurrence of outliers in a signal. Therefore, it is considered as a metric for detecting transient cavitation. The kurtosis is calculated using the following formula (the Pearson definition)

$$K_{j,k} = \frac{1}{n\sigma_{j,k}^4} \sum_{n=1}^{N_{j,k}} (x_{j,k}[n] - \overline{x}_{j,k})^4 \tag{7}$$

where $\overline{x}_{j,k}$ and $\sigma_{j,k}$ are the mean and standard deviation of the coefficients of the node (j,k).

Skewness measures the extent to which a distribution is asymmetrical. It is given by

$$Sk_{j,k} = \frac{1}{n\sigma_{j,k}^3} \sum_{n=1}^{N_{j,k}} (x_{j,k}[n] - \overline{x}_{j,k})^3 \tag{8}$$

The absolute value of the skewness of the coefficients is expected to be larger in magnitude when a signal includes transient components.

The crest factor is the ratio between the peak value and the mean of the signal, indicating how extreme the peaks are in a time series. It is obtained by

$$C_{j,k} = \frac{\max\left(|x_{j,k}[n]|\right)^2}{x_{rms}^2} \tag{9}$$

where x_{rms} is the root-mean-square of $x_{j,k}[n]$.

A feature matrix of these statistical metrics of all subbands (packets) is formed. The next step of the algorithm is to classify these subbands using their statistical features. This algorithm adopts k-means clustering, a commonly used technique for unlabeled classification [37, 38] which is discussed in the next section.

D. k-means clustering for feature classification

k-means splits a dataset comprised of n observations, each with a number of features, into k clusters. The aim of the algorithm is to minimize the total loss measured as the sum of the Euclidean distances between the centroid of the cluster and each point assigned to it (see Eq. (10)). To do this, the cluster centroids are initiated randomly and each data point is assigned to the nearest cluster. The clusters are iteratively re-allocated to minimize the loss function.

$$\Theta = \sum_{i=1}^{n} \sum_{j=1}^{k} \rho^{(i,j)} \|x^{(i)} - \mu^{(j)}\|^2 \tag{10}$$

where Θ is the loss function, ρ is the assignment of the data point i to cluster j, and x and μ are the data point and cluster centroid respectively.

k-means is a non-convex optimization problem and yields different results depending on the starting position of the clusters. This is further discussed in section II E. This effect is mitigated using multiple runs of the k-means algorithm. The clustering with the best value of the loss function is selected. Furthermore, the k-means++ algorithm [39] is used to select the starting centroids. This method selects data points as clusters one-by-one, while enforcing that points further away from existing selections are more likely to be chosen. This can improve clustering time and performance, reducing the chance that clusters are merged.

Another issue of the k-means algorithm is that the number of clusters k must be specified before training. This presents a challenge in this scenario where the number of clusters is expected to vary depending on the type of cavitation that occurs. For example, a signal comprised of a tone and noise, representing a measurement of stable cavitation, results in nodes which fall into two clusters: those representing noise, and those representing the fundamental frequency (harmonic component). Once harmonics and transients are added to the signal, more clusters could be formed. Therefore, the algorithm must adapt to the number of clusters that exist in the data. Several techniques have been developed for choosing the optimal number of clusters. [40, 41] Here, the silhouette score will use the silhouette score to to automate the selection of k. The silhouette score is a measure of how similar an observation is to its own cluster compared to other clusters, ranging from -1 to 1, where a high value indicates that the observation is well matched to its own cluster and poorly matched to neighboring clusters. An average silhouette score for all observations can be used to measure the overall quality of the clustering. The silhouette score is easy to implement and works well for small datasets. [42] It is given by

$$s = \frac{b - a}{\max(a, b)} \tag{11}$$

where b is the mean distance between a point and the points in the nearest cluster a is the mean distance between the point and all other points in the same cluster. The nearest cluster is found by finding the cluster centre that is closest the the point. k is iteratively incremented within a defined range to maximise the silhouette score.

E. Method overview

The proposed algorithm is outlined in Fig. 1. The PCD data is normalized by taking off its mean and dividing by its standard deviation. Next, the DWPT is applied to the signal, decomposing it into frequency subbands using the minimum entropy decomposition algorithm and the statistical metrics are calculated from the reconstructed data at each leaf node. The nodes with frequency higher than the PCD's cutoff are excluded. These metrics are normalized using a standard scalar, fitting each set of statistical metrics to a standard Gaussian distribution to ensure that their difference in scale does not affect clustering.

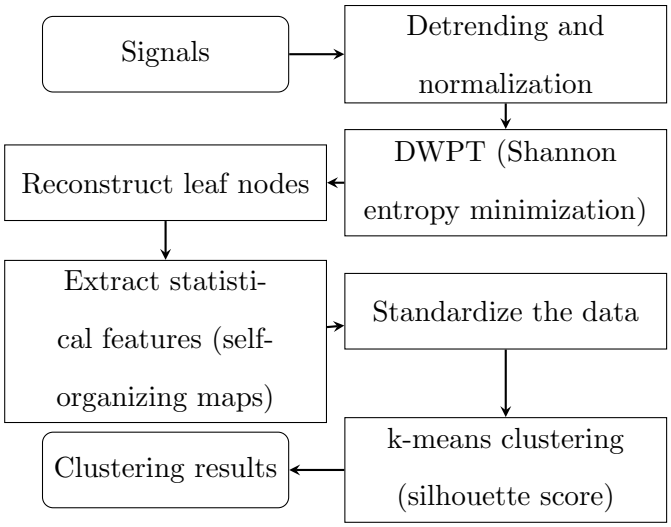


FIG. 1: Flowchart of the proposed method

These statistical metrics are fed into the k-means algorithm to for clustering. The wavelet function must be specified, however the depth of decomposition and the number of clusters are determined automatically.

The algorithm is implemented as a Python open-source library, [27] and validated using synthetic acoustic emissions generated by numerically solving Gilmore–Akulichev equations. Its performance is then tested and verified on experimental PCD data.

F. Gilmore–Akulichev model

Gilmore's equation describes the dynamics of a spherical bubble in a compressible liquid, undergoing symmetric oscillations due to acoustic excitations. The model has been validated against experimental data and widely used for studying acoustic cavitation, especially in the transient regime.[32] The Gilmore equation is expressed as follows [43]

$$R\ddot{R}\left(1 - \frac{\dot{R}}{C}\right) + \frac{3}{2}\dot{R}^2\left(1 - \frac{\dot{R}}{3C}\right) = H\left(1 + \frac{\dot{R}}{C}\right) + \frac{R\dot{H}}{C}\left(1 - \frac{\dot{R}}{C}\right) \tag{12}$$

where R is the instantaneous bubble radius, C and H are the speed of sound in the liquid at the bubble wall, and the enthalpy difference in the liquid at the bubble wall (at pressure

$P(R)$) and at infinity (at P_∞), respectively. P_∞ is the sum of the static reference pressure P_0 and the acoustic pressure $P_{ac}(t)$, i.e., $P_\infty(t) = P_0 + P_{ac}(t)$. Time derivatives are designated by dots. Gilmore's model implicitly assumes that the acoustic pressure is applied instantaneously and uniformly around the bubble without specifying a source location in the liquid. Thus, the liquid density and speed of sound at infinity are taken at the static reference state, denoted by ρ_0 and c_0, respectively, below. C and H are calculated using the following equations

$$C = \sqrt{c_0^2 + (n-1)H} \tag{13}$$

$$H = \frac{nA^{1/n}}{(n-1)\rho_0}\Big[(P(R) + B)^{(1-1/n)} - (P_\infty(t) + B)^{(1-1/n)}\Big] \tag{14}$$

where $A, B,$ and n are constants in the Tait's equation of state which are obtained experimentally. The time derivative of the enthalpy in the liquid at the bubble wall becomes

$$\dot{H} = \frac{A^{1/n}}{\rho_0}\Big[\dot{P}(R)\,(P(R) + B)^{-1/n} - \dot{P}_{ac}(t)\,(P_\infty(t) + B)^{-1/n}\Big]. \tag{15}$$

Note that $\dot{P}_\infty(t) = \dot{P}_{ac}(t)$. $P(R)$ is calculated using a modified Laplace equation

$$P(R) = \left(P_0 + \frac{2\sigma}{R_0}\right)\left(\frac{R_0}{R}\right)^{3\gamma} - \frac{2\sigma}{R} - \frac{4\mu\dot{R}}{R} \tag{16}$$

where R_0 is the initial bubble radius, γ, μ and σ are the adiabatic index (ratio of specific heats), dynamic viscosity, and surface tension, respectively. Finally, taking the time derivative of the above equation, we arrive at

$$\dot{P}(R) = -3\gamma\frac{\dot{R}}{R}\left(P_0 + \frac{2\sigma}{R_0}\right)\left(\frac{R_0}{R}\right)^{3\gamma} + \frac{\dot{R}}{R}\frac{2\sigma}{R} + 4\mu\left(\frac{\dot{R}}{R}\right)^2 - \frac{4\mu\ddot{R}}{R}. \tag{17}$$

Eqs. 13-17 are used to numerically solve the Gilmore's equation Eq. (12) . The radiated pressure in the liquid is then given by the Akulichev formula

$$P_r(t) = A\left(\frac{2}{n+1} + \frac{n-1}{n+1}\left(1+\frac{n+1}{rc_0^2}G\right)^{1/2}\right)^{2n/(n-1)} - B \tag{18}$$

where r is the distance from the centre of the bubble, and $G = R(H + 0.5\dot{R}^2)$. This equation assumes that acoustic emissions from an oscillating bubble propagate as outward traveling spherical waves. The Gilmore–Akulichev (GA) model is nondimensionalized and solved numerically using a stiff ordinary differential equation solver for different sonication conditions with the sampling rate of 100 times of the excitation frequency. The implementation is included in the GitHub repository. [27] The implementation and numerical results have been verified against published results for identical parameters. White Gaussian noise (WGN) with the signal-to-noise ratio of 30 dB was added to the calculated pressures to simulate the measurement noise.

III. RESULTS AND DISCUSSION

A. Feature Selection

The feature space is explored using self-organizing maps, [44] a type of neural network maps high dimensional data in two dimensions and is primarily used for data visualisation and clustering. The algorithm starts with predefined two-dimensional grid of nodes, each containing an array of weights, one representing each input feature. The algorithm will iteratively adjust the weights of the nodes to fit them to the features in the training data; the nodes can be thought of as cluster centres which are interconnected such that neighboring nodes have similar features. The grid of nodes is plotted for each feature, giving a set of maps which can be visually compared to evaluate the correlations between features. The details of the algorithm can be found in Haroun et al. [45], used in the context of bearing fault detection and diagnosis.

The data utilized to train the self-organizing map uses 192 signals generated with the GA model, representing both stable and transient cavitation. The signals have noise added

to them and are windowed and normalized as described in Fig. 1 - see Section III B for computational details and parameters. The results are shown in Fig. 2. When the maps are visually similar, their respective features are correlated with each other. This shows that the high energy DWPT nodes tend to have low skewness, kurtosis, and normalized spectral entropy, as expected given the physical interpretations of each. This area represents packet nodes containing fundamental frequencies. The normalized spectral entropy and level entropy are able to find nodes around the high energy area. These represent harmonics which are not easily picked out using average energy alone. The nodes with high normalized spectral entropy and kurtosis, and low energy indicate signals with sharp transients. Finally, the nodes with high normalized spectral entropy but low kurtosis and energy represent measurement noise. Depending on the statistical properties of the measurement noise, different threshold values can be defined to distinguish between the cavitation broadband component and the measurement noise. Here, the nodes are thresholded based on the kurtosis value of 3, assuming the measurement noise is white Gaussian. The metrics of the centers of the clusters are used for thresholding.

Several plots look similar, indicating redundancy in the feature space. Skewness, kurtosis, and crest factor perform similarly, while level entropy (normalized by the energy of the original signal), and normalized spectral entropy are highly correlated. Overall, the best features are hand picked as average energy, kurtosis, log average energy, and normalized spectral entropy and used for identifying nodes representing periodic, transient and noise (broadband and measurement) components.

The results for self-organizing maps are highly inconsistent where the shape of the maps can significantly change depending on the randomly-selected starting points. However, the observed correlations between the features remain consistent with each run.

B. Bubble dynamics data

The incident ultrasound pressure is modelled as a sine wave, i.e., $P_{ac}(t) = P_A sin(2\pi ft)$ where P_A and f are the amplitude and frequency of the ultrasound wave. The frequency and pulse length are set to 1.1 MHz and $100/f$, to mimic experiments, and $P_A = 1$MPa for the results presented below. We choose water and oxygen for the liquid and the gas. The physical parameters used for numerical simulations are as follows (at room temperature 293 K and

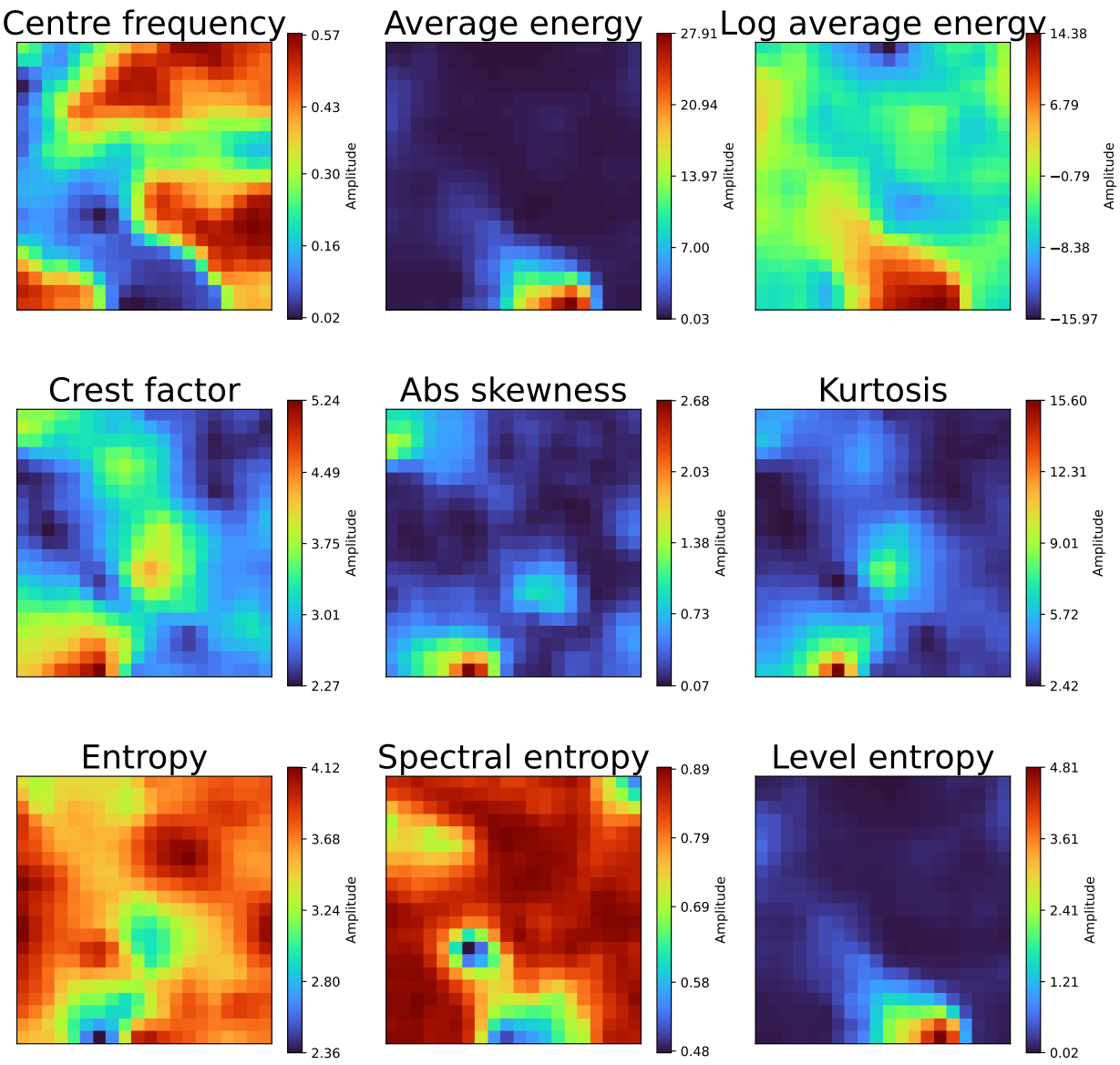
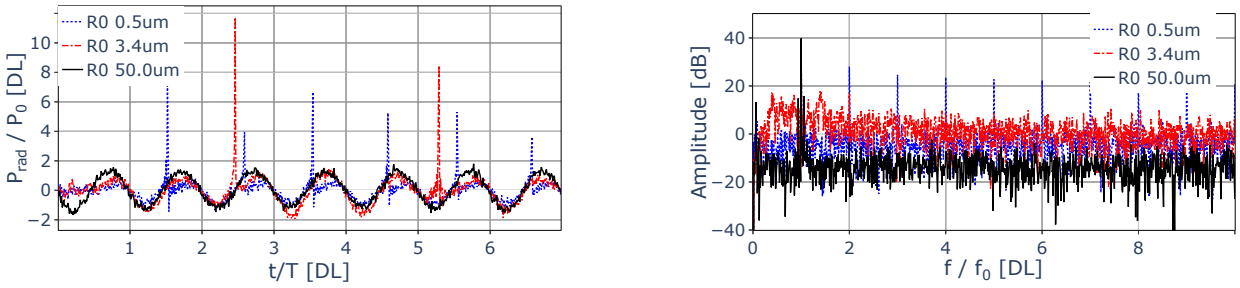


FIG. 2: Self-organizing map results are presented, with each feature plotted to illustrate its relationships with other features. The data was generated through minimum entropy decomposition of emissions, based on the Gilmore-Akulichev model. The DWPT is carried out using decomposition levels ranging from 5 to 10, and employing the db30 wavelet.

atmospheric pressure $p_0 = 100\text{kPa}$): $\rho_0 = 10^3\,\text{kg}\,\text{m}^{-3}$, $\sigma = 7.25 \times 10^{-2}\,\text{N}\,\text{m}^{-1}$, $\mu = 10^{-3}\,\text{Pa}\,\text{s}$, $c_0 = 1482\,\text{m}\,\text{s}^{-1}$, $n = 7$, $A = 3.04 \times 10^2\,\text{MPa}$, $B = 3.039 \times 10^2\,\text{MPa}$ and $\gamma = 1.4$. The initial conditions are: $R(0) = R_0$ and $\dot{R}(0) = 0$. The radius of a resonant bubble at the driving frequency (1.1 MHz) is 3.4 µm - calculated using the linear resonance equation.[30]

Here, three scenarios are simulated in which the initial radii are (i) much smaller than the resonant bubble, i.e. $R_0 = 0.5\,\mu m$, (ii) the resonant bubble radius; $R_0 = 3.4\,\mu m$, and (iii) much larger than the resonant bubble, i.e., $R_0 = 50\,\mu m$, resulting in transient and stable cavitation regimes.

The radiated pressures at $r = 100\,\mu m$ and their spectra are shown in Fig. 3.



(a) (b)

FIG. 3: Simulated acoustic emissions from bubbles with the initial radii of 0.5 µm,3.4 µm, and 50 µm; the time series (a) and spectra (b) of emissions during the first 7 cycles of excitations.

The bubble with $R_0 = 50\,\mu m$ sustains periodic stable oscillations and irradiates cyclic pressure. The main components of the spectrum are the bubble's resonance frequency at $0.06f$ and the fundamental. Considering the low energy and the kurtosis ($K = 3.5$) of the transient component, it mainly represents the added WGN (the kurtosis of the WGN is approximately 3). The $R_0 = 0.5\,\mu m$ bubble experiences cyclic transient cavitation, which is manifested as impulsive transient components in the emissions. The reconstructed transient component of the acoustic emission manifests harmonics, ultra-harmonics at $(2m+1)f/2, m \geq 2$ and broadband spectral components. In this case, the bubble is driven below its resonance which is about 11.7 MHz, thus, $f/f_r \approx 1/10$. The emissions from the resonant bubble of $R_0 = 3.4\,\mu m$, include the $3f/4$ and $f/2$ sub-harmonics, the fundamental, an ultra-harmonic at about $3f/2$, the second harmonic and broadband.

To analyze the emissions, the Daubechies wavelet function of order 30 (db30) was chosen in accordance with the physics of cavitation and the impulsive nature of desired transient component in transient cavitation. The depth of the decomposition was determined for each signal using the minimum Shannon entropy criterion, discussed before. The choice of the wavelet function is not unique. Numerical studies (not shown here) confirmed that

other orthonormal high order wavelet functions can extract cavitation signatures in the signal. Statistical metrics defined in Section II C were calculated for all leaf nodes. From the feature space analysis, a smaller subset of features — namely normalized spectral entropy, log average energy, kurtosis, and average energy — was selected for clustering and classification of the packets, as detailed in Section III A.

To classify the leaf nodes, the minimum number of clusters was set to be two and let the k-means clustering algorithm identify the clusters in the nodes using the silhouette score. The clusters are sorted from the highest average energy ($Cluster0$) to the lowest one.

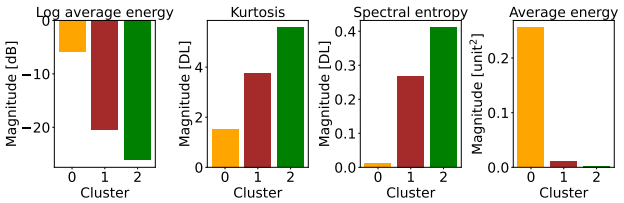
For the the submicron bubble, four clusters are identified which the first cluster ($Cluster0$) includes the periodic component as it has the highest energy and the lowest kurtosis, see Figs. 4a, 5a and 5b. $Cluster1$ includes nodes which represent the harmonics and ultra-harmonic components and $Cluster2$ includes sub-harmonics and a high-kurtosis ($K = 17.8$) broadband component. The metrics of the centers quantify different components included in each cluster.

The leaf nodes of the emissions from the 3.4 micron bubble are classified into four clusters: $Cluster0$ includes the fundamental frequency, i.e., the periodic component, see Figs. 4b, 5c and 5d. It has the highest energy and lowest kurtosis among other cluster centers. $Cluster1$ includes the sub-harmonics, the second harmonic and a low-frequency part of broadband and noise. The high-frequency broadband and noise are contained in $Clusters2$ and 3.

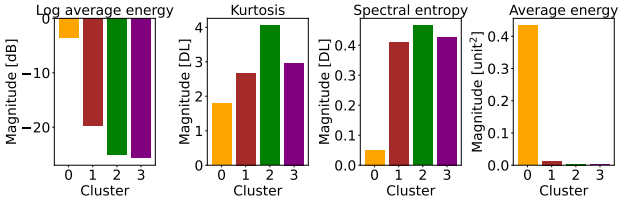
Finally, the emissions from the large 50 micron bubble are classified into four clusters, where the transients are divided into a high-kurtosis broadband component, i.e., $Cluster2$, and the low-kurtosis noise $Cluster1$, see Fig. 4c.

C. Analyzing PCD data

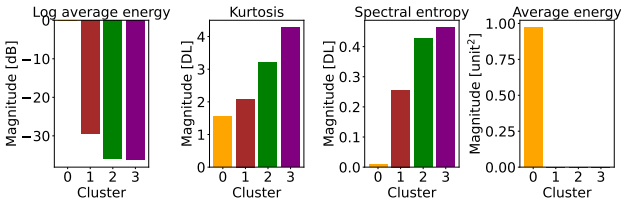
A focused ultrasound transducer (H102, Sonic Concepts, WA, USA) was driven with a burst of 100 cycles at the fundamental frequency of 1.1 MHz and different amplitudes. Focused PCDs with centre frequencies of 1 MHz, 5 MHz and 10 MHz were used to measure acoustic emissions from the focus of the ultrasound transducer. Experiments were carried out in a tank filled with de-ionized water (oxygen concentration: 7.00 ± 0.2 mg L^{-1} for all trials) at room temperature (22 °C). Refer to Haqshenas et al [26] for the detailed description of the test set-up, data acquisition parameters and experimental procedures. Here, the readings of



(a)



(b)



(c)

FIG. 4: Classification metrics of the centers of the leaf nodes clusters identified by the k-means algorithm (a) 0.5 µm, (b) 3.4 µm, and (c) 50 µm bubble.

the 1 MHz PCD are presented, as its center frequency overlaps with the excitation frequency. Two experimental scenarios were investigated, characterized by calculated peak-positive (P_{p+}) and peak-negative (P_{p-}) pressures at the focus: i) low-amplitude sonication where $P_{p+} = 3$ MPa and $P_{p-} = 2.6$ MPa, denoted as D1 below, ii) high-amplitude sonication $P_{p+} = 12.5$ MPa and $P_{p-} = 7.1$ MPa, denoted as D2. In the scenario D1, the incident pressure waves remain linear, whereas in the case D2, the waveform becomes strongly distorted, leading to the formation of higher harmonics. These harmonics can excite bubble nuclei smaller than the resonance bubble at the excitation frequency $f = 1.1$ MHz.

The wavelet transform uses minimum entropy decomposition, the db30 wavelet, symmetric padding, and enforces a minimum decomposition level of 7 to ensure that there is sufficient frequency resolution to extract closely spaced frequency components. The maximum decomposition level is determined automatically by the minimum entropy decomposition algorithm. Nodes with frequency range above a 10 MHz are excluded from analysis. The

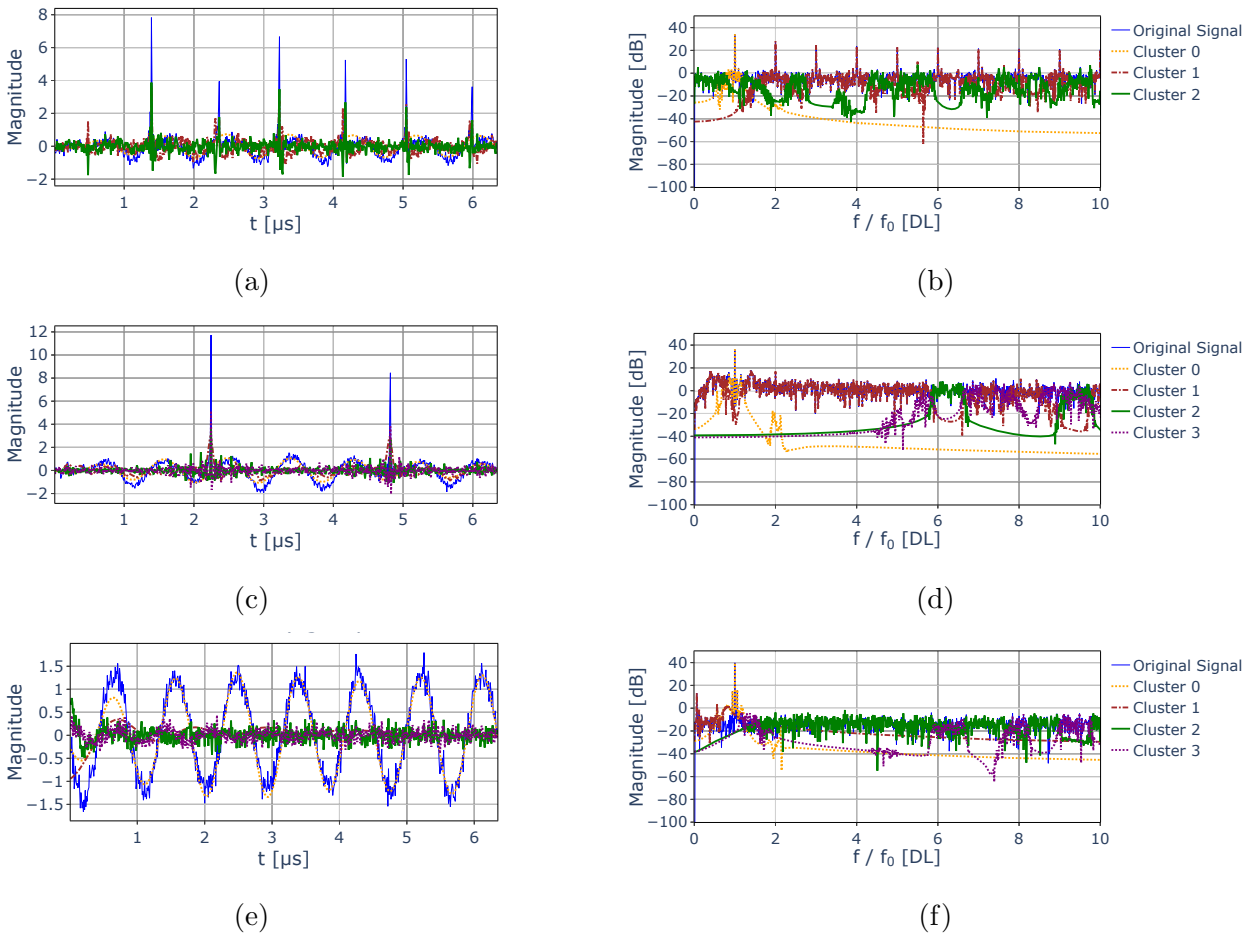
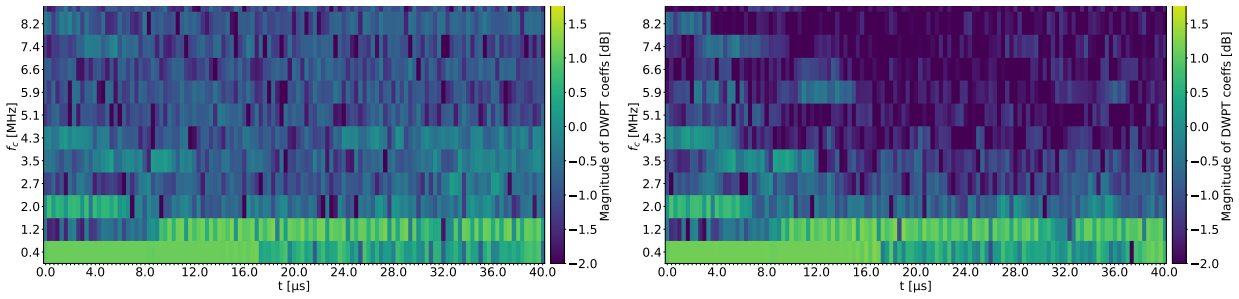


FIG. 5: Reconstructed components of acoustic emissions and their spectra, generated from bubbles with the initial radii of (a,b) 0.5 µm, (c,d) 3.4 µm, and (e,f) 50 µm.

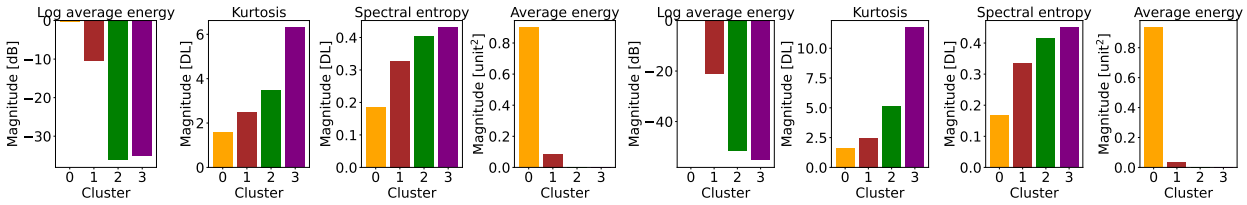
results from the wavelet transform are presented in Heisenberg plots in Fig. 6. The color of each Heisenberg box indicates the absolute amplitude of the corresponding wavelet packet coefficient. The plot reveals a higher energy band around the sub-harmonics, the fundamental frequency, and the second harmonic. A stronger broadband component above the second harmonic is evident in the D1 data. These plots also illustrate the trade-off between frequency and time resolution. The minimum entropy decomposition algorithm tends to achieve a finer frequency resolution around higher energy parts of the frequency spectrum i.e. at the fundamental frequency.

The k-means is set to select clusters based on the silhouette score, with a minimum of 3 clusters. The algorithm divides the leaf nodes of the D1 and D2 data into three and four clusters, respectively. In both cases, the fundamental is the high-energy node that has been



(a) (b)

FIG. 6: Heisenberg plot of the DWPT transform; (a) D1 data, (b) D2 data.



(a) (b)

FIG. 7: Classification metrics calculated at the centers of clusters. (a) D1 data, (b) D2 data.

singled out by the algorithm, denoted as $Cluster0$, as shown in Fig. 7.

The time series reconstructed from these clusters and their spectra are shown in Fig. 8. The clusters of packets with high kurtosis and low energy have broadband spectrum and represent the broadband components of the signals, i.e., $Cluster2$ in D1. The broadband and measurement noise are separated by the kurtosis and normalized spectral entropy values. In the low-intensity data (D1), the algorithm identifies a medium-kurtosis, low-energy and high-spectral entropy cluster ($Cluster2$). This cluster has characteristics of measurement noise. A higher-kurtosis, low-energy cluster is also identified ($Cluster3$) and has the characteristics of broadband noise.

The higher intensity ultrasound produces a prominent harmonic at $2f_0$ and a sub-harmonic at $\frac{f_0}{2}$, visible in Fig. 8d. The algorithm finds four clusters for this signal, which represent fundamental frequency ($Cluster0$), sub-harmonic, low-kurtosis low-energy noise, and the first harmonic ($Cluster1$), high-kurtosis low-energy noise ($Cluster2$) and high-frequency high-kurtosis and low-energy noise ($Cluster3$). The algorithm is able to au-

Exp.	Energy	Kurtosis	Skewness	Spec. entropy
D1	$0.37\,e-3$	18.5	0.19	0.19
D2	$6.02\,e-3$	18.5	1.19	0.60

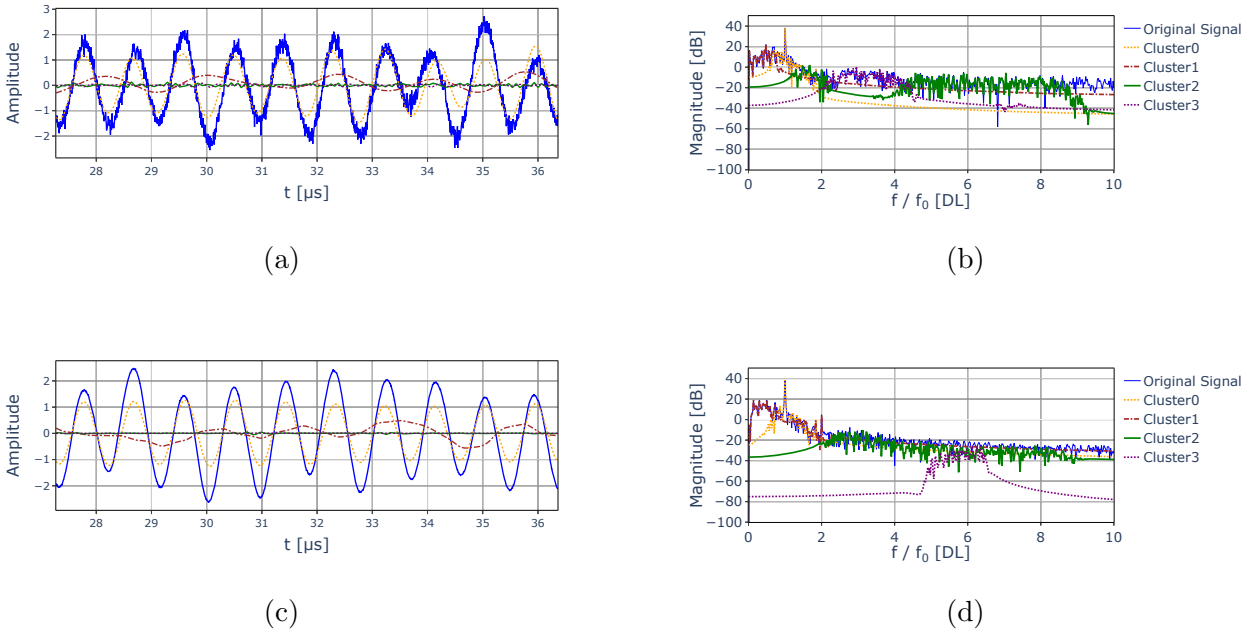
TABLE I: Statistical metrics of reconstructed data thresholded above kurtosis threshold, $K > 3$. Energy values are converted back to non-standardised form $E_i = E_{\text{std,i}} * \sigma_i^2$ where σ is the standard deviation of the data from experiment i.

Experiment	RMS	FWHM	$1/e^2$
D1	2.14-4.27	2.63-3.18	2.20-4.16
D2	1.80-4.18	2.31-3.31	1.93-3.97

TABLE II: Quantified broadband frequency ranges using frequencies above RMS, half peak, and $1/e^2$ of the spectrum. Values of frequency are normalised by the driving frequency, $f_0 = 1.1\,\text{MHz}$.

tomatically extract the fundamental frequency from the broadband and measurement noise, and harmonics but does not separate the harmonics and sub-harmonics from the noise. To quantify the broadband component, kurtosis thresholding was carried out by extracting the the clusters which have centers of kurtosis $K > 3$. Subsequently, all the nodes belonging to these clusters are searched and thresholded as well. The time series are then reconstructed using these nodes, as shown in Fig. 9a. Statistical features were calculated based on the reconstructions, shown in table I.

To find the frequency bandwidth of the broadband component, the spectra of the reconstructions are calculated and thresholded. Sections of the spectrum above the threshold are included in the frequency bandwidth. In optical applications, different methods have been used to identify bandwidths [46]. These methods have not been employed in the study of PCD data so three criteria have been tested: the RMS of the spectrum, the half of the peak $p/2$ (known as the full-width half maximum or FWHM), and p/e^2. The spectrum oscillates significantly in the thresholding range so a zero-order Gaussian filter with standard deviation is used to smooth the curve before thresholding. This method is displayed in Fig. 9b. The frequency bandwidths are shown in table II.



(a) (b)

(c) (d)

FIG. 8: Reconstructed time series of the leaf nodes of different clusters. (a) D1 data, (c) D2 data. And their spectra (b) D1, (d) D2



(a) (b)

FIG. 9: (a) Reconstructed broadband components of the D1 and D2 data sets. (b) Thresholding to define the frequency bandwidth of the broadband component for D1 data.

IV. CONCLUSION

This paper presents a novel approach for analyzing acoustic emissions measured by PCDs, utilizing multi-resolution analysis through DWPT and k-means clustering. The combination of DWPT with minimum entropy decomposition proved effective in flexibly separating the transient and periodic components of the emissions. Several metrics were evaluated to

classify the leaf nodes and quantitatively characterize cavitation features. Self-organizing maps were used to visualize the feature space and identify a subset of the most relevant metrics. A novel method for identifying frequency bandwidths of broadband components is also introduced.

The algorithm was tested on data from both numerical experiments and experimental measurements, successfully distinguishing the periodic and transient components of the emissions. It also identified and characterized cavitation features such as harmonics, sub- and ultra-harmonics, and broadband components. This was achieved by linking the physical interpretation of these metrics to the time-frequency signatures of acoustic emissions. A quantitative approach was proposed and utilized to determine the bandwidth and statistical metrics of the broadband component. This provides an effective tool for quantifying various cavitation features in PCD data and for quantitatively characterizing acoustic cavitation.

Since the statistical characteristics of the measurement noise are unknown, it is challenging to separate the measurement noise from broadband component of transient cavitation, especially when the kurtosis is low. Additionally, when leaf nodes were not sufficiently sparse, e.g., located at the boundary of two or more clusters—the k-means algorithm produced inconsistent classifications. This is mainly due to the known limitations of the k-means algorithm, such as performing hard clustering, assuming even cluster sizes and using Euclidean distances. Future work will aim to address this issue by employing alternative classification methods.

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